



DPUSNDXWavBlend

Playback of a continuous value by mixing a set of WAV-files and optionally positions the sound source in the sound simulation.

DPUSNDXWavBlend uses OpenAL for sound playback and works fine with other SNDX-DPUs, for example *DPUSNDXVstSource* and *DPUTRFSNDX*.

1. Inputs:

<i>Enable</i>	Samples will only be played if this input is not 0. default: 1
<i>Volume</i>	Volume of playback Range is 0 (silence) .. 1 (maximum). default: 1
<i>VolStep</i>	Volume is changed by this value at maximum (per simulation step). Low values lead to a smoother transition but may add a delay if the input value changes suddenly. default: 0.05
<i>Threshold</i>	Range (around a sample position value, see below) where only one sample is played. Higher values improve the reaction time but may result in unclear interpolation. default: 0.05
<i>In</i>	Value to be used (for example velocity)

2. Inputs for 3D positioning:

<i>PositionMode</i>	Mode of the positioning. 0 (default): no positioning, standard stereo playback 1: absolute positioning in the SILAB world 2: relative positioning to listener
<i>X, Y, Z</i>	Position of the sound source. Either absolute (Position-Mode=1) or relative (PositionMode=2) to listener.
<i>vX, vY, vZ</i>	Velocity of the sound source. Absolute to the SILAB world coordinate axes (PositionMode=1) or relative to the listener (PositionMode=2)

3. Parameters:

<i>SoundPath</i>	Directory in which the samples are located. The path must be end with an backslash „\“ (see functional principle). If no absolute path is set, the directory is searched within the SILAB install path „lib\sounds“ and within the project path
<i>Files</i>	Filenames of the sound samples (see functional principle).
<i>Values</i>	Corresponding input values for each sound sample (see functional principle).



4. Functional principle:

This DPU is used to mix WAV-files according to a continuous value (*In*). A set of sound samples recorded at discrete values of *In* is a precondition. The DPU mixes two adjacent samples if the value *In* is between two sample values.

All samples must be applicable for endless play, i.e. loop-able.

Parameters of the DPU consist of the tuples for the files and the corresponding values. Both tuples have to be the same size. An empty item in the files-tuple may be used to use no sound for the corresponding value.

5. Example:

```
DPUSNDXWavBlend Rain {  
    Computer = {SNDX};  
    Index = 50;  
  
    SoundPath = "rain/";  
  
    Values = (0, 0.2,          0.7,          1);  
    Files = ("", "RainLow.wav", "RainMedium.wav", "RainHigh.wav");  
  
    PositionMode = 2;  
    X = 0;  
    Z = 1;  
};  
Connections = {  
    Environment.OutPrecipitation -> Rain.In  
};
```

This example shows a high quality simulation of raindrops hitting the top of the car.

Three sound samples are used (*RainLow.wav*, *RainMedium.wav* and *RainHigh.wav*) (see Files-tuple) and the corresponding values are set in the Values-tuple: at a rain density of 0 no sound is played (= „empty file“), at 0.2 *RainLow.wav* is played, and so on. At a rain density of 0.5 *DPUSNDXWavBlend* mixes the samples *RainLow.wav* and *RainMedium.wav* proportionately which results in a continuous increase of the sound.

The samples are loaded into RAM before the start of the simulation. Using more than one instance and playback of this DPU will cause the sounds to be mixed. Depending on the computers hardware there might be side effects like high latency.

Positioning:

Optionally the sound source can be positioned in the 3D environment.

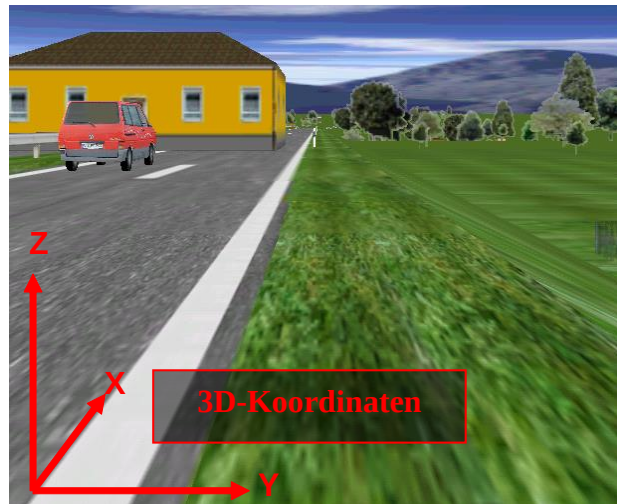
Note: Only mono channel WAV-files will be positioned correctly. Stereo or multichannel files force PositionMode=0.



SILAB DRIVING SIMULATION

version Version 7.1 documentation

Coordinates of the sound source are specified in the SILAB world coordinate system. The origin of this coordinate system is the setup point of the car during startup. At this time positive X-values point to the front, positive Y-values to the right and positive Z-values up. The vectors unit is meters. Positive angles (rad) along the (positive) axis are counter-clockwise.



Using **PositionMode = 1** absolute coordinates for the sound source from this origin are expected. The velocity along the axes is needed to calculate any Doppler effect and is expected in meters per second.